

The Commutator as Measurement Object

Depth, Persistence, Cost, and Transparency in a Change-First Algebra

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Abstract

We develop a constructive account of discrete mechanics in which the sole primitive is the mandate to change, formalized as the requirement that every executable state possess a distinct legal successor. From this single constraint, together with the difference operator, we show that a balanced shift dynamics (a de Bruijn attractor) is generated rather than assumed, and that the natural state of such a dynamics is the edge rather than the vertex. On the edge, evolution is bijective; the projection back to the vertex is the sole site of information loss. This reproduces, in miniature, the fold/hash split familiar from cryptographic compression. Within this setting we treat the commutator of two transformations not as a scalar but as a measurement object with several independent projections. Two of these projections — the order of the commutator and its fixed-point count — turn out to be genuinely distinct functionals: the first measures the settlement cost of reversing an interaction (which we identify with inertial mass), the second measures how much of the state space survives the interaction (which we identify with transparency, and its complement, opacity). We prove computationally that these two channels can disagree sharply, refuting the tempting identification of mass with opacity. We then decompose the accumulated depth of a composite transformation into an axial part (built inside a commuting family) and a cross part (built across non-commuting families), and show that axial depth accumulates coherently while cross depth thermalizes to a fixed statistical background. A control experiment with matched cycle-type statistics but different commutation graphs establishes that this coherence is a property of the algebraic topology, not of the individual generators. The result is an order parameter that turns "structure" from a descriptive label into a measurable phase variable, monotone in persistence, mass anisotropy, and directional transparency. Throughout, every load-bearing claim is confirmed by direct computation, and several intuitively appealing hypotheses are recorded as killed.

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1. Introduction

The program reported here begins from an unusually austere starting point and attempts to see how much of the structure of discrete mechanics can be forced from it. The starting point is a single constraint, which we will call C_1 and state informally as: all things must change. This is not a slogan but an operational requirement, and much of the value of the present work lies in making it precise enough to have consequences. In its sharpest form, C_1 says that in the dynamics of a system, every executable state must have a distinct legal successor. A state that maps to itself has stopped executing; it has no continuation, no difference, no history to accrue. Under C_1 such a state is not merely unlikely but excluded from the domain of the dynamics.

The second primitive is the difference operator, which we write as Δ . Difference is what change looks like when you measure it: the claim that something has changed is the claim that a difference exists between before and after. In a binary setting the difference operator is exclusive-or, the modulo-two difference, and this identification will let us connect the abstract constraint to concrete, executable transformations.

From these two primitives we recover a surprising amount. We recover a canonical balanced dynamics — the de Bruijn shift — not as an imposed symmetry but as the unique object that survives the constraints. We recover the distinction between the reversible core of a computation and its irreversible projection, the same distinction that appears in the cryptographic literature as the difference between a keyed permutation and its feed-forward compression. We recover a natural hierarchy of derived observables — velocity, mass, acceleration — as successive reads of a single lifted state. And, in the portion of the work that this paper treats most carefully, we recover a principled account of what it means for a composite object to have structure, opacity, inertia, and persistence, and we show that these are not four independent properties but several projections of one underlying object: the commutator.

The methodological commitment throughout is that the compiler is the arbiter. No structural claim enters the record without a live computation confirming it. This discipline has repeatedly paid off. Several claims that seemed clean on paper were falsified when run, and in each case the falsification handed back a sharper structure than the one it destroyed. We report the killed claims alongside the surviving ones, because the pattern of failures is itself informative: it maps the boundary between what the framework forces and what merely happened to be true of a particular ensemble.

The paper is organized as follows. Section 2 states the constraint engine precisely and shows how the balanced shift is generated. Section 3 develops the edge lift and the fold/hash split. Section 4 introduces the commutator as a measurement object and separates its projections. Section 5 treats mass as the settlement channel and establishes its scaling laws, including the singular regime that appears when a transformation is its own inverse. Section 6 treats transparency and opacity as the fixed-point channel and proves their independence from mass. Section 7 develops the axial/cross decomposition of depth and the order parameter for structure, and reports the control experiment that locates the mechanism in the commutation topology. Section 8 collects the killed claims and the open questions. Section 9 concludes.

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1.1 On method

A word about the style of reasoning is in order, because it differs from the usual pattern in speculative work and the difference is the point. The usual pattern is idea, then interpretation, then a supporting example. The pattern here is prediction, then a live computation, then acceptance or rejection of the prediction by the result of that computation, then, in the case of rejection, a repair and a rerun. The computation is not decoration on an argument settled in advance; it is the arbiter, and it is allowed to overrule the argument. Several of the strongest results in the paper are strong precisely because a prior, weaker or wrong version was run and found wanting. Replacing the claim that equal change forces the balanced shift with the claim that difference plus the change-mandate forces it, and retracting the universality of the involution coefficient in favor of an ensemble-dependent calibration, are two examples where accepting a correction produced a more defensible statement than the one it replaced.

This method has a cost and a benefit. The cost is that the paper carries a ledger of killed claims, which a more polished presentation would suppress. The benefit is that the surviving claims have all passed a falsification attempt, and the reader can see where the boundary lies between what the framework forces and what merely happened to hold in a particular ensemble. We regard the ledger as an asset rather than an embarrassment, and Section 8 collects it deliberately.

1.2 The constraint hierarchy in brief

It will help to have the dependency structure in view before the details. The primitive is the change-mandate, C_1 , read as continuability: every executable state has a distinct successor. Composed with the difference operator, C_1 selects a maximal-novelty recurrence, whose single fixed point at the origin it then forbids, forcing a unique minimal repair that yields the balanced shift. The shift's reversible dynamics lives on the edge state, and the projection to the vertex is the sole information sink. On the edge sits the jet tower of derived observables. Interaction among transformations is measured by the commutator, whose order and fixed-point count are two independent channels carrying, respectively, inertial mass and transparency. Accumulated depth splits into a coherent axial part and a thermalizing cross part, and the axial fraction is an order parameter for structure whose mechanism resides in the commutation topology. Each arrow in this chain either constructs the next object or forces a correction that preserves a prior constraint; none imports an outside axiom. The remainder of the paper walks this chain and substantiates each link with computation.

2. The Constraint Engine

2.1 C_1 as continuability

It is worth dwelling on the exact content of C_1 , because the loose reading ("things move") is weaker and less useful than the sharp one. The sharp reading is a statement about the execution rule of a system rather than about any particular object within it. Let the dynamics be a map from a state space to itself. C_1 requires that this map have no fixed point: for every state, the image differs from the state. Equivalently, every executable

state must admit a distinct legal continuation. The name we prefer for this is continuability. The dynamics must always be able to take another step that is genuinely a step.

This framing immediately dissolves a tension that would otherwise trouble the framework. If C1 forbids stasis, how can there be stable structures at all? The resolution is that C1 constrains the law, not the cargo. The execution rule may not fix any state; but an object inside the system — an element of the transformation algebra, a participant rather than the dynamics itself — is under no such prohibition. It may fix a great many states. We will see in Section 6 that an object may fix all but three of the states in the space and still be a legal participant. The engine may not idle; the things the engine moves may be as sluggish as they like, short of total immobility.

2.2 Difference and the maximal recurrence

Take the difference operator over a binary alphabet. The simplest deterministic dynamics built from it is a linear recurrence: the next symbol is a fixed exclusive-or combination of previous symbols. This is a linear-feedback shift register in the language of coding theory, but the point to hold onto is that it is nothing more than a discrete difference equation, the modulo-two analogue of a linear recurrence over the reals.

Among such recurrences, C1 selects a particular one through the drive toward maximal novelty. A repeat adds nothing new; a system obeying C1 should, whenever it can, move to a state it has not yet visited. Formalize novelty as the count of distinct states visited up to a given time, and require that this count strictly increase whenever possible. Under a finite state space and deterministic evolution, maximizing novelty is equivalent to maximizing the length of the orbit before it closes. For a linear difference recurrence, the maximal-orbit choice is the one whose characteristic polynomial is primitive, and its orbit is the maximal-length sequence: it visits every nonzero state exactly once before returning.

We verified that the greedy novelty walk — at each step, take an unvisited window whenever one exists — visits all of the states with strictly increasing novelty at every tested width. This is Martin's classical result on de Bruijn sequences, recovered here as the direct consequence of the novelty principle rather than imported from register theory. The novelty count increases by exactly one at every step until saturation, and the walk never stalls before visiting everything.

2.3 The forced repair and the birth of the shift

The maximal linear recurrence has one defect, and it is exactly a C1 violation. The all-zero state maps to itself. Under a linear rule with no constant term, zero is a fixed point: it does not change. This is the unique obstruction to C1 in the entire construction, and it sits at the origin.

C1 forbids it, and the minimal repair that removes it is forced. To make the zero state change, its feedback bit must flip; there is only one value it can flip to, so the zero state is compelled to map to a specific nonzero state. This flip, however, breaks the bijectivity of the map at the target: two states now map to the same place. To restore bijectivity — which is required if the dynamics is to be reversible, a property we will need in Section 3 — the feedback of exactly one other state must flip, and there is only one state whose flip repairs the collision. We verified by exhaustive search over all possible flip sets that the minimal repair has size two, that it is unique

at every tested width, and that it consists precisely of the origin and its algebraic opposite. Within the class of difference recurrences, the repair is not merely unique up to relabelling; it is forced at every step, with no free choice anywhere.

The repaired dynamics is a single orbit through every state — a de Bruijn shift. Its transition graph has uniform in-degree; every legal transition appears exactly once. In earlier framings we had assumed uniform transition measure as a separate axiom ("all change is equal") and derived the shift from it. The corrected and stronger statement is that uniform measure is not assumed at all: it emerges as a consequence of the difference operator meeting C1 and repairing the single fixed point. The symmetry we once posited at the top of the framework is now exhaust from the bottom.

The interpretive payoff is worth stating plainly. The origin is the one place where change stops, the unique degenerate state. Everything the framework builds downstream — the shift, the reversible core, the derived observables — is the resolution of that single collision between the mandate to change and the one state that refuses to. The universe, in this reading, is the exhaust of the void failing to hold still.

3. The Edge Lift and the Fold/Hash Split

3.1 Vertices are the wrong state

Consider the dynamics as a walk on a graph whose vertices are the states and whose directed edges are the legal transitions. If a vertex can be reached by more than one predecessor, then asking "what comes next after this vertex" is ill-posed: the answer depends on how you arrived. This apparent branching is not a real indeterminacy in the dynamics. It is an artifact of asking the question at the wrong granularity.

The fix is to lift the state from the vertex to the edge: the true state is not the current position but the ordered pair of the previous position and the current one. On the edge, the dynamics is single-valued and reversible. The pair (previous, current) evolves to (current, next), and because the current position together with its predecessor determines the transition uniquely, this evolution is a bijection. The apparent fork at a convergent vertex disappears once the predecessor coordinate is restored. We confirmed this on the balanced shift: every recurrent vertex is reached by exactly two predecessors, so the vertex-level successor is genuinely two-valued; but the edge-level evolution is a single cycle through all edges, and its inverse exists and composes back to the identity exactly.

3.2 The projection is the writer

There are thus two maps in play. The first is the edge evolution, which is bijective and loses nothing: it is the reader. The second is the projection from the edge back to the vertex, which forgets the predecessor coordinate. This projection is many-to-one; it destroys exactly the information needed to distinguish the histories that converged. It is the writer, and it is the only place in the entire round where information dies.

This is the same structural decomposition that appears in cryptographic compression. A keyed permutation is invertible — a fold that remembers everything and can be retraced. The compression step, which adds the

original input back to the permuted output, is the sole non-injective operation, and it is what makes the whole construction one-way. We verified this correspondence directly on a full implementation of a standard hash compression function: its rounds, given the message schedule, form a bijection whose inverse can be read off by running the rounds backward, recovered without error over thousands of random states; the single non-injective step is the feed-forward addition, and everything else in the round preserves information. The fold is the reader on the edge; the hash is the writer at the projection. Nothing else destroys information.

The accounting of the erasure is exact and worth making explicit, because it identifies where entropy is produced in a dynamics that is otherwise entirely reversible. On the balanced shift, the edge carries one more coordinate than the vertex — the predecessor bit — and the projection discards exactly that coordinate. Each application of the projection therefore destroys precisely one bit of information, no more and no less, and it destroys it at exactly the convergent vertices where two histories meet. Everywhere else the dynamics is a bijection and conserves information perfectly. The second law, in this miniature, is not a statistical tendency but a bookkeeping identity: entropy is produced at a fixed rate of one bit per projection, at the merges and nowhere else. The reversible fold is the conservative core; the projection is the dissipative surface; and the two together are the whole of the dynamics. This is the sharpest form we can give to the intuition that irreversibility is not fundamental but is manufactured at a specific structural site — the merge — while the underlying evolution remains reversible.

A further consequence deserves note. Because the edge evolution is a single cycle through all edges, and because reading a single coordinate off a point in that cycle is a well-defined map, the two diagnostic operations we have studied elsewhere — the bijective read that addresses a position in a reversible structure, and the many-to-one read that merges positions into a lattice — are the same operation distinguished only by whether the read is one-to-one or many-to-one. The reversible read is the edge dynamics; the merging read is the projection. They are not two different mechanisms but two cardinality-cases of one addressing operation, which is why the fold and the hash share so much structure despite their opposite reversibility.

3.3 The jet tower

Once the edge is the primitive state, position as a bare quantity recedes and the natural objects are successive lifts. The vertex is the zeroth lift. The edge — the pair of consecutive positions — is the first lift, and it carries velocity, the first difference. The triple of three consecutive positions is the second lift, and it carries acceleration, the second difference. These are the finite jets of the trajectory, and the observables we care about are simply reads of the tower at different heights.

This reorganizes what would otherwise look like a collection of unrelated mechanical quantities into a single hierarchy. Velocity is a first-jet read. Acceleration is a second-jet read. And, as we will argue in Section 5, the inertial mass of a composite object is also a first-jet property in the sense that it is defined on the edge state and not on the bare vertex. Force, inertia, and acceleration are not distinct kinds of thing that need to be related to one another; they are projections of one lifted state, read at successive orders. On the balanced shift we confirmed the sharp version of this: the second difference is single-valued on every edge and multi-valued on every vertex, so acceleration is a function on edges and merely a relation on vertices — exactly the

same enrichment that mass requires. Mass and acceleration are the same kind of object because they live on the same enriched state.

4. The Commutator as a Measurement Object

4.1 Interaction is non-commutation

Let two transformations act on the state space. If applying them in one order gives the same result as applying them in the other, they do not interact: their commutator is the identity. The commutator is defined as the composite of the first, the second, the inverse of the first, and the inverse of the second. If this composite is the identity, the two transformations commute and the interaction bill is zero. If it is not the identity, the composite fails to return to the starting configuration, and the residue it leaves is the trace of the interaction.

The central move of this paper is to stop treating the commutator as a single number. The residue it leaves has internal structure, and different features of that structure encode different physical quantities. We identify at least two independent projections of the commutator, and the burden of Sections 5 and 6 is to show that they are genuinely independent — that knowing one does not determine the other.

4.2 The two channels

The first projection is the order of the commutator: the number of times it must be applied before it closes back to the identity. This is a temporal-depth measure. It counts how many applications of the interaction residue are needed before the configuration returns to where it started. We will identify the logarithm of this order with the settlement cost of the interaction, and ultimately with inertial mass.

The second projection is the fixed-point count of the commutator: the number of states that the residue leaves exactly where they were. This is a visibility measure. It counts how much of the state space passes through the interaction unaltered. Many fixed points means most of the space survives — the interaction is nearly transparent. Few or no fixed points means almost nothing survives — the interaction is opaque. We will identify a normalized version of this with transparency and its complement with opacity.

These are different questions about the same operator. Order asks how expensive the return path is; fixed points ask how much survives each application. There is no a priori reason they should track one another, and the empirical work below shows that they do not. The clearest discriminator is a residue that completely rearranges the state space — no fixed points at all — yet closes after only two applications. Such an object is maximally opaque and minimally deep. Its existence, which we exhibit in Section 6, is the proof that opacity is not inertia.

4.3 The symmetry of the encounter

Before developing the two channels quantitatively, we record a structural fact that dissolves an old paradox and clarifies what the commutator is a measurement of. Consider the venerable puzzle of the unstoppable force meeting the immovable object. In the present algebra, an unstoppable force would be a transformation that passes through every object without leaving a residue — one that commutes with everything. An

immovable object would be a transformation unmoved by every other — again, one that commutes with everything. These are the same condition, read from the two sides of the interaction: membership in the center of the transformation group.

We computed the center of the group generated by the full set of elementary transformations on a state space of a thousand and twenty-four elements. It contains exactly one element: the identity. The only unstoppable force is the transformation that does nothing; the only immovable object is likewise the transformation that does nothing; and these are the same object, which is precisely the degenerate non-execution that C₁ excludes. The paradox was never a genuine contradiction. It arose from taking one commuting relation and splitting it, with language, into two contradictory nouns — a force and an object — and then demanding that the two halves fight. The algebra never distinguished them. We verified directly that the commutator of A with B is the exact inverse of the commutator of B with A: identical in magnitude, identical in structure, opposite in reading. Which operand is called the force and which the object is a decision made by the observer, not a property carried by the pair. The interaction is one thing; the two names are ours.

This is why the commutator is the right measurement object. It measures the encounter, not either party. Its order and its fixed-point count are properties of the interaction as a whole, symmetric under exchange of the two participants up to inversion, and the physical quantities we extract are therefore properties of relationships rather than of isolated things.

4.4 The absolute is off the manifold

One further computation sharpens the status of the excluded absolute. If the transformation group contained a graded approach to stasis — elements commuting with ninety percent of the space, then ninety-nine percent, and so on — then C₁ would be a soft boundary, statistically bypassable. We measured the global commuting fraction of every real element against a large sample of deep transformations. The identity sits at unity: it commutes with everything. Every non-identity element sits at exactly zero: it commutes with essentially none of the deep transformations. The gap is total and unbridged. There is no sequence of increasingly immovable objects converging on immovability. The absolute is not the limit of a long road of increasingly stable things; it is disconnected from the state space, with no address on the manifold at all. C₁ is not a speed limit that could in principle be approached. It is the absence of the road.

Locally, by contrast, stability is abundant. Real elements commute with a few percent of their immediate neighborhood — enough to build recognizable structure, never enough to freeze. Reality, in this picture, is negotiated: every real object carries both persistence and change, enough of the former to be recognized and enough of the latter to satisfy C₁. These are not competing principles but the same execution seen at two scales.

5. The Settlement Channel: Mass

5.1 Mass is not displacement

The first serious attempt to locate inertial mass in this algebra failed, and the failure was instructive. The natural guess is that mass is displacement resistance: a heavy body, pushed, should move less. But every elementary transformation in the algebra is a thorough mixer. Applying any non-trivial transformation to a state scrambles roughly half of its bits, and composing a few transformations saturates this scrambling almost immediately. We measured the displacement residue of the commutator of a body with a push and found it pinned at the thermal floor — half the bits flipped — by the time the body had any appreciable depth, with essentially no variation across bodies. In the displacement channel there is no gradient, and therefore no such thing as a heavy body. A feather and an anvil scramble to the same thermal equilibrium.

This is a genuine kill. Mass-as-displacement is refuted. But the manner of the refutation points directly at the correct channel. If every interaction saturates the displacement metric, then mass cannot live in the amount of scrambling; it must live in the structure of the return path. This is the order of the commutator — the settlement channel.

5.2 Mass is settlement depth

Define the mass of a body, relative to a push, as the logarithm of the order of the commutator of the body with that push: the number of settlement steps, in bits, required to close the interaction loop. Unlike displacement, this quantity has a gradient. Across many bodies of the same construction depth, the settlement mass varies substantially, with a spread-to-mean ratio two orders of magnitude larger than the displacement channel could support. And unlike displacement, this quantity is center-independent: the order of a permutation does not depend on which state you start from, so mass here is an intrinsic property of the body alone, not a relational accident of the state it happens to act on. That is exactly what one wants of mass. It is a property of the thing, not of where the thing is.

The settlement mass accumulates with the depth of the body up to a ceiling. A body built from a few transformations has modest mass; as more transformations are wrapped around it, the mass climbs, and then it saturates. The saturation is not a failure of the measurement but a real feature: a finite state space can only support so much settlement depth. The ceiling is set by the configurational capacity of the space, and we return to its scaling below.

5.3 The two regimes and the selector bit

The scaling of the settlement ceiling with the size of the space turns out to depend on a single algebraic property of the push, and this is one of the sharper results in the program. Generically — when the push is a transformation distinct from its own inverse — the settlement order scales as the square of the logarithm of the space size. This is the typical order of a random permutation, the regime governed by the classical theory of the order of random permutations, and it represents a kind of thermal ceiling: the interaction wanders the full space and the return path is as long as a random return path can be.

But when the push is its own inverse — an involution, a transformation of order two — the dynamics leaves this regime entirely and drops to a scaling as the square root of the space size. The transition is triggered by the single condition that the push equals its own inverse. A directed push rides the square-of-logarithm ceiling; a directionless push, one that reciprocates rather than circulates, drops to the square-root line. One bit of information about the push — whether it is its own inverse — selects between two fundamentally different asymptotic laws.

The interpretive reading of this selector deserves to be stated, because it connects the mechanical result to the change-mandate that anchors the framework. A transformation distinct from its own inverse has a direction: applying it and applying its inverse are different acts, a circulation with an orientation. A transformation equal to its own inverse has no direction: applying it twice returns to the start, a reciprocation with no orientation. The generic regime is the regime of directed motion; the singular regime is the regime of directionless motion. The condition that a push equals its own inverse is exactly the condition that the push carries no direction, and it is precisely at that condition — and only there — that the two settlement observables diverge from one another and the dynamics leaves the thermal ceiling for the square-root line. Direction, which one might have been tempted to posit as an independent axiom of the mechanics, is here revealed as the selector between two regimes, with its absence marking the singular point. The mechanics does not assume direction; it exhibits the one directionless push as its lone singularity.

We traced the mechanism to its root. The involution push, at the relevant magnitude, is the half-swap: the transformation that exchanges the two halves of a word. As a permutation of the states it has a definite fixed-point set, namely the states whose two halves are equal, and there are exactly the square root of the space-size many of these. The commutator of a body with this involution is then the product of two involutions — the half-swap and its conjugate by the body — each fixing a square-root-sized diagonal. And here the deeper structural theorem enters.

It is worth pausing on why the half-swap in particular carries this square-root-sized fixed set, because the number is not a coincidence. A word is fixed by the half-swap exactly when its two halves are equal. The number of such words is the number of possible half-words, which is the square root of the number of possible full words. So the fixed set of the half-swap is the diagonal of the space, and the diagonal of a product space has exactly the square root of the product's size. The square root that governs the singular regime is thus, at bottom, the size of the diagonal of a space that factors into two equal halves — a geometric fact about the shape of the state space, surfacing as a scaling law of the dynamics.

5.4 The fragmentation theorem

The controlling result behind the square-root regime has nothing to do with orders or logarithms or return paths. It is purely combinatorial and geometric. Consider the product of two involutions, each fixing some number of points. We tested, across a range of that number and across space sizes, the following law: the product has approximately that many cycles, and its mean cycle length is the space size divided by that number. The measured ratios converge cleanly to unity from above as the number of fixed points grows, with only a small correction at small counts. The fixed-point count is the controlling order parameter, not the space size.

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The consequence is immediate. When the number of fixed points is the square root of the space size, the product has a square root of the space size many cycles, each of mean length the square root of the space size. The square-root fragmentation is therefore forced by the fixed-point count of the involution, before any question of orders or least common multiples is even posed. The half-swap fixes a square-root-sized diagonal; two such involutions multiplied together fragment the space into square-root-many cycles of square-root length; and the settlement order inherits that square-root scaling. This is a topological boundary, not a statistical artifact, and it explains why the involution line exists at all.

5.5 The coefficient and its status

Having established the scaling, one can ask for the constant in front. The settlement order on the involution line is proportional to the square root of the space size, and the proportionality constant, measured across four space sizes spanning a sixty-four-fold range, converges to a value near one and two-thirds, stable within error and free of drift once past the smallest space. We decomposed this constant into a base contribution from small divisors — those below the square root of the space size, which the bulk of the cycle lengths cluster near — and a tail contribution from large divisors, reached only by the longest cycles, which extend several times past the square-root scale before petering out.

Here we exercise the honest ledger. We initially described this constant as a universal constant of products of involutions. A control experiment refuted the universality claim. Varying the ensemble — comparing bodies built from the elementary transformation alphabet against uniformly random involutions with the same number of fixed points, and varying the construction depth — moved the constant at the second decimal place: uniformly random involutions sit slightly higher than the alphabet-built bodies, and under-mixed shallow bodies sit noticeably lower. The constant is therefore an empirical coefficient of a sufficiently mixed ensemble, not a universal invariant of the abstract product-of-involutions problem. What is universal, and proven, is the square-root scaling itself, which follows from the fragmentation theorem independent of any coefficient. The engine is the scaling; the coefficient is its calibration, and its value carries a dependence on the ensemble that we now state explicitly rather than paper over.

6. The Fixed-Point Channel: Transparency and Opacity

6.1 Transparency is commutation

We now develop the second projection of the commutator and show that it is independent of the first. Define the transparency of a push relative to a body as the fraction of states on which their commutator acts as the identity — the fraction that passes through the interaction unaltered. Full transparency is complete commutation: the push slides through the body without a trace. Zero transparency is a completely opaque interaction: nothing passes through unchanged.

Measured across the elementary transformations, transparency shows exactly the graded structure one would hope for. Same-axis pairs — a rotation against another rotation, a multiplication against a negation — are fully transparent: they commute completely. Cross-family pairs — a rotation against an addition, an

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exclusive-or against a multiplication — are fully opaque: they collide on every state. And between these extremes there is genuine partial transmission: a rotation passes through a multiplication on a small but nonzero fraction of states, the residue of their near-alignment. No transformation is transparent to everything, and none is opaque to everything, which is the center-triviality result of Section 4 reappearing as a spectrum: there is no universal ray and no universal wall.

This gives a physical reading of a familiar phenomenon. That some things pass through other things — that a ray penetrates a medium — is, in this language, the statement that the ray's transformation nearly commutes with the medium's, so most of the medium's state space is left unaltered by their interaction. Transparency is not a substance but a relation, and it is graded by how nearly the two transformations commute.

6.2 Opacity is not inertia

We can now settle a hypothesis that had been tempting throughout the program: that opacity and inertia are the same quantity, both being manifestations of accumulated non-commuting history. The intuition is seductive. A body is hard to turn because of its internal cross-term bill; a body is hard to see through because of the same accumulated interaction structure; surely these are one thing.

They are not, and the commutator-as-measurement-object framing shows exactly why. Mass is the order of the commutator; opacity is the fixed-point deficit of the same commutator. These are different functionals of the same operator, and they can disagree. The decisive counterexample is a body-and-push pair whose commutator is a fixed-point-free involution: it has order two, so its settlement mass is a single bit — almost free to reverse — yet it has no fixed points, so it is completely opaque. Almost weightless and completely opaque, in the same interaction. We exhibited exactly such a pair: a rotation-built body against an exclusive-or push, whose commutator turns nearly the whole space over but closes after two applications.

The correct statement, therefore, is not that opacity is inertia but that opacity and inertia are two independent reads of one underlying object. The commutator is confirmed as the single common source; mass reads its order and opacity reads its fixed points; and because these are different features of the residue, the depth dial that drives both does not force them to be equal. This is a cleaner and stronger position than the identity hypothesis, and it is the position the data supports.

6.3 Persistence splits by role

The fixed-point channel also resolves the apparent tension between the mandate to change and the existence of persistent structures, and it does so by distinguishing two notions of persistence that had been entangled. The first is dynamical persistence, a property of the law: the execution rule may have no fixed point, on pain of freezing the system. This is the C_1 requirement, and it is exactly what forces the fixed-point repair that generates the shift. The second is structural persistence, a property of objects: an element of the algebra, being a participant rather than the dynamics, may fix as many states as the group structure allows.

We asked whether there was a ceiling on structural persistence and initially conjectured, on the strength of the half-swap's square-root-sized fixed set, that the square root of the space size was the maximum. A directed search refuted this decisively. Driving the support of an element downward through iterated

commutators of near-commuters, we found elements fixing all but three of the states — vastly more than the square root. The search then stalled at exactly three moved points, and the stall is not a limitation of the search but an exact algebraic bound. A non-identity bijection cannot move one point, and cannot move only two while remaining an even permutation; since the elementary transformations generate the even permutations, the minimum support of a non-identity element is three. This gives a concrete quantum of change: the least a legal transformation can do and still be doing something is to move exactly three states. The maximum structural persistence is achieved by the minimum legal change.

This kills the square-root persistence conjecture cleanly, and the balancing expansion is more valuable than the conjecture. Persistence does not have a single ceiling; it has two, according to role. As a dynamics, an element may fix nothing. As an object, an element may fix all but three. Confusing these two ceilings is what made the square-root scale look like a persistence limit, when in fact the square-root scale belongs to the involution fragmentation regime and has nothing to do with maximum persistence. The half-swap fixes a square-root-sized set not because that is the persistence ceiling but because that is the characteristic fragmentation scale of order-two dynamics — an entirely different statement.

7. Depth Decomposition and the Structure Order Parameter

7.1 The axial/cross decomposition

We arrive at the central constructive result of the paper: a decomposition of accumulated depth that explains what it means for an object to have structure. The depth of a composite transformation is the number of elementary transformations wrapped around it. But not all depth is alike. Depth accumulated inside a commuting family — where each added transformation commutes with those already present — behaves entirely differently from depth accumulated across non-commuting families.

We measured this directly by tracking the fixed-point count of composite transformations as a function of depth, separately for words built from a single commuting family and words built from the full mixed alphabet. The mixed words thermalize: their fixed-point count falls to the statistical background of a random permutation — approximately one fixed point on average, the Poisson expectation — and stays there regardless of further depth. History, accumulated across non-commuting directions, erases persistence down to chance. This we confirmed with a clean measurement, excluding words that happened to collapse to the identity and averaging over hundreds of samples at each depth: the mixed-family average descends to the Poisson background and holds.

The commuting-family words do not thermalize. Built from a family whose members all commute, they never fall to the background; they hold a substantial fixed-point count at every depth. And in the case of a multiplicative family, the fixed-point count does not merely resist decay but grows with depth: deeper words in that family fix more states, not fewer, because composing within the family drives the element toward a highly persistent limit rather than toward randomness. Depth accumulated along a commuting direction is constructive; depth accumulated across directions is thermal. This is the mechanism of structure. A structured

object is history accumulated coherently along a commuting axis; a thermalized object is history accumulated incoherently across many axes.

7.2 Structure as anisotropic mass

The axial/cross decomposition has an immediate mechanical consequence: structure is anisotropic mass. A body built along a single axis is weightless and transparent with respect to pushes along that axis, and heavy and opaque with respect to every other. We measured the settlement mass and transparency of axially-built bodies against pushes along and across their axis. A rotation-built body, pushed by a rotation, has zero mass and complete transparency: it is free to turn along its own axis and the push passes through it untouched. The same body, pushed by a multiplication, carries full mass and near-zero transparency: it resists and it occludes. A multiplication-built body mirrors this exactly, free along its own axis and costly across it. And a mixed-built body pays in every direction — heavy and opaque to every push, with no free axis at all.

This is the difference between a crystal and a rock, falling out of the commutation structure with no additional assumption. An ordered object is cheap to move in one direction and expensive in all others; a disordered object is uniformly expensive. Order and disorder are the anisotropy and isotropy of the settlement bill. Mass, in a structured object, is not a scalar but a directional quantity, and its anisotropy is precisely the coherence of the object's internal history.

7.3 The order parameter

These observations invite a single order parameter. Let the depth of a composite object be split into an axial part — depth accumulated within a commuting family — and a cross part — depth introduced by non-commuting interactions. Define the structure parameter as the axial fraction of total depth: the ratio of axial depth to the sum of axial and cross depth. When this parameter is near one, the object is crystal-like, coherent, ordered. When it is near zero, the object is isotropic and thermal. Intermediate values describe composite structures with partial order.

We swept this parameter by building bodies from a controlled mixture of an axial generator and a cross generator, varying the fraction, and measuring persistence, directional mass, and directional transparency at each mixture. The mappings are monotone in the expected directions. As the axial fraction rises, the fixed-point count rises — more persistence. As the axial fraction rises, the mass to the object's own axis falls and its transparency to that axis rises — the object becomes ever freer and clearer along its coherent direction. The cross-axis mass stays high throughout. Structure, quantified this way, is no longer a descriptive label. It is a measurable phase variable that predicts persistence, mass anisotropy, and directional transparency together.

7.4 The control: structure lives in the topology

There remained one boundary condition to settle, and it is the sharpest test in this section. Does axial depth create structure because commuting composition genuinely preserves information, or merely because the particular generators we chose happen to carry a hidden bias? The generators in the commuting family are specific transformations; perhaps their individual identities, not their mutual commutation, are responsible for the coherence.

We ran the control. We constructed two generator ensembles with matched cycle-type statistics — each generator in one ensemble has an exact cycle-type twin in the other, so their individual orders and fixed-point counts are identical — but with deliberately different commutation graphs. The first ensemble was built to commute: all of its generator pairs commute. The second was built by relabelling each generator through an independent random conjugation, which preserves each generator's cycle type exactly while destroying the shared structure that made them commute; none of its generator pairs commute. Same individual statistics on both sides; opposite commutation topology.

The persistence trajectories split completely. The commuting ensemble holds a substantial fixed-point count at every depth out to well past a hundred compositions. The non-commuting ensemble collapses to the Poisson background within a handful of compositions and stays there. Because the per-generator statistics are identical across the two ensembles, the divergence cannot be a property of the individual generators. It is a property of the commutation graph. Structure lives in the algebraic topology, not in the symbols. The mechanism is the geometry of which transformations commute with which, not the internal identity of any transformation taken alone.

This closes the boundary condition the decomposition rested on. The fold from generators to structure passes through the commutation topology as its essential middle term: the individual generators fix the raw material, the commutation graph fixes the depth trajectory, and the depth trajectory fixes mass, opacity, and persistence. Remove the topology and the same generators produce only thermal noise; supply the right topology and they produce coherent, persistent, anisotropic structure.

8. Numerical Protocols and Representative Measurements

Because the compiler is the arbiter throughout, we set out here the shape of the computations and the representative numbers they returned, so that the claims of the preceding sections can be weighed against their evidence rather than taken on assertion. We describe protocols rather than reproduce raw tables, and we give figures to the precision that the runs support.

8.1 The transformation alphabet and the state space

The elementary transformations act on binary words of a fixed width, treated as the integers of a finite modular ring. The alphabet comprises an exclusive-or by a constant, an addition by a constant, a rotation by a fixed amount, a half-swap that exchanges the two halves of a word, a multiplication by an odd constant, and a negation. Each is a bijection of the state space, and each carries its own inverse: the exclusive-or and the half-swap and the negation are their own inverses, the addition inverts by subtraction, the rotation by counter-rotation, the multiplication by the modular inverse of its constant. This self-inverse structure is not incidental; it is what makes the never-written inverse of a composite available for free, by reading the word of transformations backward with each letter replaced by its partner.

The width was varied across experiments according to what each measurement required. Center-independent quantities such as the settlement order were measured on widths from twelve up to eighteen and, for the

scaling checks, twenty, giving state spaces from a few thousand up to a million states. Quantities requiring exhaustive enumeration of the transformation group, such as the center computation, were run on a width of ten, a thousand and twenty-four states. Cycle-decomposition measurements ran comfortably to a million states because their cost is linear in the space size.

8.2 The settlement channel

Settlement mass was measured as the base-two logarithm of the order of the commutator of a body with a push, where the order is the least common multiple of the cycle lengths of the commutator permutation, computed by cycle decomposition followed by prime-power accumulation to avoid overflow on the enormous least common multiples that arise. On the generic line, the settlement order scales as the square of the width, with a fitted exponent within a few percent of two and a coefficient near the value predicted by the classical theory of random-permutation order, the exact match being softened by the fact that a body built from a bounded number of alphabet transformations is not a uniformly random permutation and undershoots the random-permutation typical order by a stable factor.

On the involution line, the settlement order scales as the square root of the space size, with a coefficient measured across four space sizes that lands near one and two-thirds and shows no drift once past the smallest space. The base contribution from small divisors sits near one and one-tenth of the square-root scale, and the tail contribution from large divisors near five-tenths, both stable across the space sizes. The longest cycles on the involution line extend to several times the square-root scale — a factor that grows slowly with width — and it is these long cycles that reach the large divisors responsible for the tail lift.

8.3 The fragmentation theorem

The fragmentation measurement generated pairs of random involutions with an exactly controlled number of fixed points, formed their product, and counted its cycles and their mean length. The number of fixed points was swept across a wide range at two space sizes. The ratio of cycle count to fixed-point count converged to unity from above — from ratios well above one at small fixed-point counts down to within a fraction of a percent of one at large counts — and the product of mean cycle length with fixed-point count converged to the space size on the same schedule. This is the evidence that the fixed-point count, not the space size, is the controlling order parameter, and that the square-root fragmentation follows from setting the fixed-point count to the square root of the space size.

8.4 The center and the transparency spectrum

The center computation enumerated, for the group generated by the full alphabet on a thousand and twenty-four states, every permutation commuting with all generators; the enumeration is tractable because a transitive group's centralizing element is determined by the image of a single point. The result was a center of size one, containing only the identity. Restricting the generating set produced nontrivial centralizers — a set of four for a two-generator abelian-flavored subset, and intransitive behavior with surviving immovable sectors for still smaller sets — establishing that the triviality of the center is a property of the complete alphabet and not an artifact of the method.

The transparency spectrum was measured as the fraction of states fixed by the commutator of each generator pair. Same-axis pairs returned exactly one; cross-family pairs returned exactly zero; near-aligned pairs returned small positive fractions, on the order of a percent or less, quantifying partial transmission. The complementarity between transparency and displacement was measured on composite bodies of increasing depth: as the transparency of a body to a fixed push fell toward zero, the displacement residue of the same interaction rose to the thermal ceiling of half the width, the two channels behaving as complements.

8.5 Persistence versus depth and the structure control

Persistence was measured as the fixed-point count of composite transformations, averaged over hundreds of samples at each construction depth with identity collapses excluded. Mixed-alphabet words descended to the Poisson background of approximately one fixed point and held there at all depths beyond a handful. Single-commuting-family words did not descend; the rotation family held several fixed points at all depths, and the multiplicative family increased its fixed-point count with depth, reaching many times the background at the deepest words tested.

The structure control constructed two generator ensembles with matched cycle types but different commutation graphs — one built to commute on every generator pair, the other built by independent random relabelling of each generator so that no pair commutes while every generator retains its exact cycle type. The persistence trajectories diverged completely: the commuting ensemble held a substantial fixed-point count out past a hundred compositions, while the non-commuting ensemble collapsed to the Poisson background within a handful of compositions. Because the two ensembles have identical per-generator statistics, the divergence isolates the commutation graph as the cause. The order-parameter sweep, mixing an axial generator with a cross generator in controlled proportion, returned monotone dependence of persistence, own-axis mass, and own-axis transparency on the axial fraction, in the directions reported in Section 7.

9. Killed Claims and Open Questions

The honest ledger is a load-bearing part of this program, and we collect here both the claims that were refuted and the questions that remain open, because the shape of the boundary between forced and incidental results is itself a finding.

8.1 Killed claims

Mass as displacement was killed. Every elementary transformation is a thermal mixer, so the displacement channel saturates and carries no gradient; mass had to move to the settlement channel. The balancing expansion was a center-independent, intrinsic mass with a genuine gradient.

The strict interpretation "all change is equal forces the balanced shift" was killed as an over-strong implication. Uniform transition measure is one realization of equal change, not the only one; the balanced shift is instead forced by the difference operator meeting C1 and repairing the origin, with uniform measure emerging as a consequence. This reordering removed an assumed symmetry from the top of the framework and replaced it with a construction from the bottom.

The claim that the involution-line coefficient is a universal constant of products of involutions was killed by a control across ensembles. The coefficient depends on the ensemble at the second decimal place. What survives is the square-root scaling, which is proven and ensemble-independent, and the coefficient is now correctly labeled as an empirical calibration of a sufficiently mixed ensemble.

The conjecture that the square root of the space size is the maximum structural persistence was killed by a search that found elements fixing all but three states. The balancing expansion was the exact quantum of change — minimum support three — and the recognition that persistence splits by role, with the square-root scale belonging to the involution fragmentation regime rather than to persistence.

The hypothesis that opacity is inertia was killed by a fixed-point-free involution commutator: order two but no fixed points, hence nearly weightless yet fully opaque. The balancing expansion was the recognition that mass and opacity are two independent functionals of one commutator — its order and its fixed-point count — rather than the same quantity.

8.2 Open questions

Several genuine questions remain, and they are now sharp rather than diffuse. The fragmentation theorem — that a product of two involutions each fixing a given number of points has approximately that many cycles of mean length the space size divided by that number — is supported by convincing numerics and by a matching-union heuristic, but a rigorous asymptotic proof, including the correction term that governs the approach from above, is not yet in hand. That correction term appears to scale logarithmically in the space size relative to the fixed-point count, and pinning it would explain the finite-size drift precisely.

The involution-line coefficient, now correctly understood as ensemble-dependent, still lacks a derivation of its value for any specific limiting ensemble. Expressing it as an integral over the limiting cycle-length density of the uniformly random square-root-fixed involution product would turn the measured number into a computed one and explain why the random ensemble sits slightly above the alphabet-built ensemble.

The order parameter for structure invites a full phase diagram. We have established monotonicity of persistence, mass anisotropy, and directional transparency in the axial fraction, but the functional forms of these mappings, and whether any of them exhibit sharp transitions rather than smooth crossovers, remain to be characterized.

The probe-compatibility principle deserves formal statement. Across this program, every measurement channel that went flat did so because the probe nearly commuted with the observable — the probe was transparent to the structure it was meant to measure. Displacement was transparent to mass; symbol entropy was transparent to acceleration; composition statistics were transparent to path identity. In each case the fix was to find a probe that fails to commute with the structure. This suggests a general operational rule: when a measurement repeatedly returns null, ask whether the probe lies in or near the commuting algebra of the observable, and if so, choose a probe that collides. The principle is broader than the present framework and resembles ideas from quantum measurement, but here it has an exact algebraic content — the sensitivity of a measurement vanishes as the commutator of probe and observable approaches the identity — and formalizing it precisely is an open and, we think, valuable task.

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10. Conclusion

We set out to see how much of discrete mechanics can be forced from a single constraint — that every executable state must have a distinct legal successor — together with the difference operator. The answer is a good deal more than one might expect. The constraint, meeting difference and repairing its single fixed point at the origin, generates a balanced shift dynamics as an attractor rather than an axiom. The reversibility of that dynamics lives on the edge, not the vertex, and the projection from edge to vertex is the unique site of information loss, reproducing the fold/hash split of cryptographic compression. On the edge, the derived observables organize into a jet tower, with velocity, mass, and acceleration as successive reads of one lifted state.

The heart of the paper is the recognition that the commutator of two transformations is not a scalar but a measurement object with several independent projections. Its order is the settlement channel and carries inertial mass; its fixed-point count is the visibility channel and carries transparency and opacity. These channels are genuinely independent, as a fixed-point-free involution commutator demonstrates by being nearly weightless yet fully opaque. And the paradox of the unstoppable force and the immovable object dissolves into a syntactic artifact: both names refer to the same excluded absence — commutation with everything, which only the identity achieves — split by language into two nouns that were never distinct in the algebra.

The constructive climax is the decomposition of accumulated depth into an axial part, which accumulates coherently, and a cross part, which thermalizes to a statistical background. This decomposition explains what structure is: coherent history along a commuting axis, manifesting as anisotropic mass, directional transparency, and resistant persistence. A control experiment with matched generator statistics but different commutation graphs locates the mechanism unambiguously in the algebraic topology rather than in the individual generators. Structure thereby becomes a measurable phase variable rather than a descriptive label, with an order parameter monotone in persistence, mass anisotropy, and transparency.

Throughout, the discipline of confirming every structural claim by live computation, and of recording the killed claims alongside the survivors, has repeatedly converted appealing intuitions into either theorems or instructive failures. The framework that results rests on a narrow and explicit foundation: a constraint on execution, a difference operator, and the commutation topology of the transformations that the two of them generate. From that foundation, persistence, cost, transparency, and structure emerge not as separate postulates but as projections and accumulations of a single interaction object. The engine is the constraint; the observables are its shadows; and the geometry that ties them together is the geometry of what commutes with what.

We close by naming the character of what remains, because it differs from the character of what has been done. The work reported here is largely constructive and largely finished: the constraint chain is assembled, the commutator's channels are separated, the structure order parameter is defined and its mechanism localized. What remains is largely analytic. The fragmentation theorem wants a proof of its asymptotics and its correction term; the involution coefficient wants a derivation from a limiting density; the structure order

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parameter wants a full phase diagram; the probe-compatibility principle wants a formal statement. None of these is a search for new structure — the structure is in hand — and all of them are asymptotic or analytic questions about objects already constructed and already measured. That is a comfortable place for a program to be: past the phase where one does not know what the objects are, into the phase where one proves theorems about objects one can already compute. The transition from the first phase to the second is, in our view, the main thing that has happened, and it happened not despite the killed claims but through them, each refutation narrowing the foundation and sharpening the questions that survive. A change-first algebra, taken seriously and held to its computations, turns out to have quite a lot to say about why things are heavy, why things are opaque, why things persist, and why some things are ordered and others are not — and to say it in the single language of what fails to commute with what.